

AI Workbench Control Plane

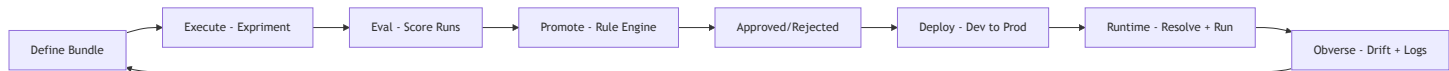
Deep Technical Architecture

Core Principle

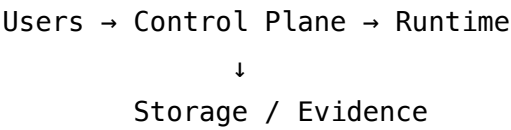
Bundle = model + prompt + retrieval + policy + evaluation

- Deployable unit is NOT the model
- Full configuration is versioned + governed

Lifecycle management = register + excute + evaluation + promote + deploy + observe (compare w/ challenger)



End-to-End Architecture



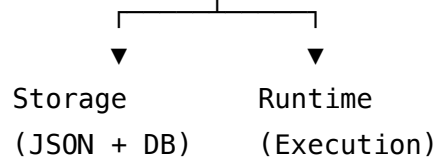
Detailed Architecture

Users (CLI / Notebook / UI)

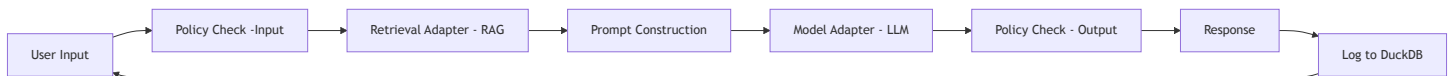


AI Workbench Control Plane

- Bundle Manager
- Experiment Runner
- Evaluation Engine
- Promotion Engine
- Deployment State Manager



RAG + Policy Runtime Flow



Runtime Resolver

env → deployments.json → bundle_id



Resolved Bundle Snapshot



Adapters (model + retrieval + policy)

- Runtime NEVER accepts bundle input

- Always resolves from registry

Bundle Lifecycle

draft → evaluated → candidate → approved → deployed

- No skipping evaluation
- No direct deployment

Lifecycle Flow

Define → Execute → Evaluate → Promote → Deploy → Runtime → Observe

Experiment Execution

- Load bundle
- Execute against eval set
- Capture:
 - inputs
 - outputs
 - timings

Artifacts:

- config.json
- predictions.jsonl

Evaluation Engine

Metrics:

- correctness
- groundedness
- policy compliance
- latency
- completeness

Outputs:

- metrics.json
- [scorecard.md](#)

Promotion Engine

metrics + rules + guardrails → decision

- must_pass rules
- regression checks
- weighted metrics

Deployment State

dev → bundle_v2
prod → bundle_v1

- Explicit environment mapping
- No "latest"

Runtime Execution Detail



Evidence Layer

Stored:

- bundle registry
- run artifacts
- metrics
- promotion decisions
- deployment history
- inference logs

Governance Loop

Runtime Logs → Sample → Evaluate → Drift Check → Alert

- Detect regression
- Compare vs baseline

Champion vs Challenger

Modes:

- Shadow
- Traffic split

Outputs:

- agreement rate
- latency delta
- policy differences

Rollback

prod: bundle_v3 → rollback → bundle_v2

- Uses deployment history
- Immediate runtime switch

Key Design Principles

1. Promote bundles, not models
2. Evaluation before deployment
3. Environment state is explicit
4. Evidence is persistent
5. Runtime resolves from registry

Final Takeaway

- Control plane governs lifecycle
- Runtime executes approved bundles only
- Full auditability and reproducibility